

■ Junior to Intermediate Developer 6-Month Roadmap

Month 1 — Solidify Core Foundations

- Deepen C#.NET Core skills: async/await, LINQ, IEnumerable vs IQueryable, memory management.
- Database mastery: indexing, joins, query optimization, EF Core advanced queries.
- Practice: Profile an existing app and optimize queries & API performance.

Month 2 — Middleware & Custom Pipelines

- Build custom middleware (logging, error handling, request timing).
- Understand ASP.NET Core pipeline and middleware order.
- Practice: Create middleware that logs request time and saves to a log file.

Month 3 — Authentication & Authorization

- Master JWT authentication: access/refresh tokens, expiration handling.
- Learn OAuth2 & OpenID Connect integrations.
- Implement role-based & policy-based authorization.
- Practice: Build an app with JWT, refresh tokens, and role-based dashboard.

Month 4 — Architecture & Design Patterns

- Study Clean, Onion, and Layered Architecture.
- Learn design patterns: Repository, Unit of Work, Factory, Strategy, Mediator.
- Practice: Refactor a project into Clean Architecture.

Month 5 — Advanced Topics & Deployment

- Learn cloud basics (Azure hosting, Azure SQL, Blob storage).
- Set up CI/CD with GitHub Actions or Azure DevOps.
- Implement caching (in-memory, Redis) and performance optimization.
- Practice: Deploy booking system with CI/CD.

Month 6 — Real-World Simulation & Mentorship

- Follow Git Flow branching strategy & perform code reviews.
- Mentor a junior dev & document your code.
- Final Project: Vehicle Tracking & Booking System v2 with all skills combined.